

# vSET Startup Screening Dashboard

Project brief: backend-driven dashboard for company screening reports

**Goal.** Turn each company's screening report (a PDF plus a matching JSON file) into a professional B2B dashboard. Report headings become sidebar tabs and the content under each heading fills that tab. Data is stored in a backend database and fetched by the frontend; nothing is hard-coded. Two companies exist today (TerraSpark and Mysa); the design must scale to many more without code changes.

## 1. What the source material contains

Both PDFs follow the same report template (14 and 15 pages). Both JSON files share one schema (vset.screen.v2), so the same code can render either company. Each PDF is the content and layout reference; each JSON holds the same report as structured data.

|                   | TerraSpark                       | Mysa                                          |
|-------------------|----------------------------------|-----------------------------------------------|
| Sector / stage    | Deeptech, space solar / Pre-Seed | Fintech, AI finance operations / Pre-Series A |
| Base country      | Luxembourg (founded 2025)        | India (founded 2023)                          |
| Funding on record | EUR 5m Pre-Seed (Mar 2026)       | USD 2.8m Seed + USD 3.4m Pre-Series A         |
| Founders          | 3 (CEO, two co-founders)         | 3 co-founders + 2 further executives          |
| Research cut-off  | 30 March 2026                    | 3 February 2026                               |
| Report reference  | f2a8fd552eed                     | 46c13f529a25                                  |

## 2. Tab map: PDF heading to JSON source

| # | Sidebar tab                                 | JSON source                                    | Content                                                                                   |
|---|---------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1 | Key facts & context                         | sections[key=company]                          | Summary, current position, profile grid, products, timeline, target markets, legal status |
| 2 | Founder & team                              | sections[key=team]                             | Founder cards with market-fit notes, team-market fit, management coverage                 |
| 3 | Product & technology                        | sections[key=product]                          | Portfolio, value chain, differentiation, maturity, core technology, IP                    |
| 4 | Validation & market signals                 | sections[key=validation]                       | Validation position, traction, next milestone                                             |
| 5 | Market opportunity                          | sections[key=market]                           | Definition, buyer and ICP, demand drivers, barriers, sizing, analyst view                 |
| 6 | Competitive landscape                       | sections[key=competition]                      | Alternatives table, comparison, defensibility                                             |
| 7 | Funding history                             | sections[key=funding]                          | Financing position, timeline, investors                                                   |
| 8 | Investor questions & information to prepare | action_requirements + actions                  | Concerns, Part A questions, Part B documents                                              |
| 9 | About this screen & sources                 | final.report_basis + evidence_register.sources | Scope, limitations, research window, source list                                          |

Each section page in the PDF ends with an "Information to prepare" list. It comes from actions[kind=SECTION\_REQUEST], matched to the section by its title.

## 3. Data findings that shape the design

- **Blocks are typed arrays**, not objects: [type, title, ...payload]. Types found: para, kv, olist, list, table, cards, comparison. A block-renderer registry maps each type to one component.

- **The two companies differ in structure.** Mysa has a key-management table in Founder & team; TerraSpark has an external-validation table in Validation. TerraSpark's comparison is a criteria list (empty header); Mysa's is a multi-column matrix with a note. Renderers must handle both comparison shapes.
- **Gaps are normal.** Cells contain "-" or "Not established" where public evidence is missing. Show a muted "Not available" state, never a blank or an error.
- **Section order and titles come from the data,** so a new section appears as a new tab with no code change. Unknown block types need a safe fallback.
- **The JSON carries more than the PDF shows** (for example 122 evidence claims in TerraSpark and 168 in Mysa, each with an evidence state). This is out of scope now, but the data model keeps the raw JSON so it can be added later.

## 4. Architecture

The build follows **SOLID** and **Clean Architecture**. Dependencies point inward only: presentation and infrastructure depend on application, application depends on domain, and domain depends on nothing.

| Principle             | How it shows up in this project                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| Single responsibility | Separate classes for parsing, validating, persisting, mapping and rendering.                             |
| Open / closed         | New block types and sections are added by registering a renderer or mapper; existing code is not edited. |
| Liskov substitution   | Every block renderer takes the same props contract and can replace any other.                            |
| Interface segregation | Small ports such as ReportReader and ReportWriter, kept separate.                                        |
| Dependency inversion  | Use cases depend on repository interfaces; Prisma and HTTP are adapters injected from outside.           |

### Suggested stack (changeable with a stated reason)

- **Backend:** Node.js, TypeScript, NestJS, PostgreSQL (JSONB), Prisma, Zod, Jest, Docker Compose.
- **Frontend:** React, TypeScript, Vite, React Router, TanStack Query, Tailwind CSS, Vitest.

### Backend structure

```

backend/src/
  domain/          company, report, section, block, source, action-item
  application/     use-cases/ (ListCompanies, GetReportHeader, GetSection,
                    GetActions, GetSources, ImportReport)
                    ports/ (ReportRepository, CompanyRepository, Clock)
                    mappers/ (raw JSON -> domain)
  infrastructure/  persistence/prisma/  ingestion/ (schema check, import)
  presentation/    http/ (controllers, DTOs, error filter)

```

### Frontend structure

```

frontend/src/
  app/             providers, router, layout shell
  core/            http client, env, error types
  shared/          ui/ (Button, Chip, Card, Callout, DataTable, KeyValueGrid,
                    Skeleton, EmptyState, ErrorState) lib/ styles/
  features/        companies/ (switcher)
                    report/ domain/ data/ ui/ (shell, blocks + registry,
                    sections, actions, sources)
  routes/

```

## 5. Data model and API

**Model.** Company (1 to many) Report (audience, version, generated\_at, research\_cutoff, fingerprint, raw JSON snapshot); Report (1 to many) Section (key, title, position, ribbon, blocks as JSONB); plus Source and ActionItem (kind, where, text, why, group, priority or secondary). Import is idempotent, keyed on canonical\_screen\_id with final\_fingerprint for change detection.

| Endpoint                                  | Returns                                                   |
|-------------------------------------------|-----------------------------------------------------------|
| GET /api/v1/companies                     | Company list for the switcher                             |
| GET /api/v1/companies/:slug               | Header: cover fields, status ribbon, audience label       |
| GET /api/v1/companies/:slug/sections      | Navigation: key, title, position                          |
| GET /api/v1/companies/:slug/sections/:key | Ribbon, blocks and the section's "information to prepare" |
| GET /api/v1/companies/:slug/actions       | Part A questions by topic; Part B documents by group      |
| GET /api/v1/companies/:slug/sources       | Source list                                               |
| POST /api/v1/reports/import               | Admin only; validates and stores a raw report JSON        |

## 6. UI specification

- **Look:** B2B, professional, clean. Light theme, neutral slate, one navy accent taken from the PDFs, thin dividers, no gradients or emoji, Inter or system font.
- **Layout:** collapsible left sidebar (about 260 px) with the nine tabs; top bar with wordmark, company switcher and report meta; report header with cover fields and status chips.
- **Routing:** deep links at /companies/:slug/:sectionKey; back button works.
- **States:** skeleton, empty and error-with-retry for every data view.
- **Accessibility and responsive:** WCAG AA contrast, keyboard navigation, aria-current on the active tab, drawer sidebar on narrow screens.

## 7. Delivery phases and acceptance criteria

| Phase | Work                                                          | Done when                                             |
|-------|---------------------------------------------------------------|-------------------------------------------------------|
| 1     | Implementation plan                                           | You have reviewed and approved it                     |
| 2     | Backend foundation: schema, entities, importer, seed          | Both JSON files import; re-import changes nothing     |
| 3     | API, use cases, OpenAPI, tests                                | All endpoints return all nine tabs for both companies |
| 4     | Frontend foundation: shell, tokens, primitives, routing       | Sidebar, switcher and empty pages work                |
| 5     | Report views: block registry, section pages, actions, sources | Every PDF page's content appears in the right tab     |
| 6     | Polish: states, accessibility, responsive, PDF comparison     | Side-by-side check against both PDFs passes           |

**Scalability test.** Adding a third company must need no code change: import its JSON and it appears in the switcher with its own tabs. A contract test with a small variant JSON should prove this.

## 8. Assumptions and open questions

- Authentication and roles are out of scope now; a guard seam is left for later.
- The audience is "Founder" (from meta.audience). Other audiences may exist later, so the model stores audience per report.
- Which paragraph titles appear as highlighted callouts is not marked in the data. The prompt lists the ones seen in the PDFs; keep that list in one config file and confirm it during phase 6.
- Confirm with your company: the wordmark "vSET", brand colours, and whether the import endpoint should be exposed or only run as a CLI.